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A Spatio-Temporal Constrained Anti-Interference Correlation Filtering Target Tracking Algorithm

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Abstract—Trackers based on correlation filtering (CF) can balance accuracy and efficiency simultaneously, and have the advantages of low cost and low data dependence, thus being widely applied in fields such as intelligent monitoring. However, in the practical application of complex scenarios, it is confronted with severe challenges such as low resolution and drastic changes in appearance. To solve this problem, we propose a spatio-temporal constrained anti-interference visual target tracking algorithm. Firstly, introduce the spatial gradient regularization term in the spatial constraint module to constrain the spatial smoothness of the tracking weights and suppress the interference of background noise. Secondly, by pre-detecting the response value map of the previous frame image, the background area information similar to the target is extracted and added to the training of the filter to improve the discrimination ability of the filter. Finally, the alternate direction multiplier method is used to optimize and solve the filter. The optimization problem is decomposed into three sub-problems, and the global optimal solution is approximated through iterative updates. Experimental tests were conducted on the public dataset OTB50, OTB100 and UAV-123. The experimental results show that the algorithm proposed in this paper exhibits higher robustness in scenarios such as severe target occlusion, deformation, and illumination changes, and has

better tracking performance.

Keywords—Correlation filtering, Spatio-temporal collaboration, Complex scenarios, Anti-interference.

I. INTRODUCTION

Visual object tracking is one of the core and popular research directions in the field of computer vision. This technology has extensive practical applications in multiple fields, including unmanned aerial vehicles [1], item detection on production lines [2], autonomous driving [3], intelligent transportation [4], military operations [5], and human-computer interaction [6], etc. It has significant research significance.

In the field of target tracking, significant progress has been made in methods based on CF, among which the STRCF[7] algorithm demonstrates outstanding tracking capabilities. However, when confronted with challenging scenarios such as severe occlusion, complex background interference or target rotation, this algorithm still has problems such as loss of tracking targets and scale estimation bias. The performance degradation of STRCF in complex environments mainly results from the following two factors: First, the spatial distribution of the algorithm's tracking weights is not constrained, making it difficult to suppress the spatial heterogeneous interference of background noise and leading to a decrease in positioning accuracy; Second, the historical frame information is not fully

utilized, making it difficult to distinguish the target from similar distractors (such as obstructions or similar objects in the background), which can easily lead to tracking drift.

To address the above issues, based on the STRCF algorithm framework, we propose a spatio-temporal constrained anti-interference correlation filtering target tracking algorithm.

Our main contributions are as follows:

- Introduce a spatial gradient regularization term into the objective function of the filtering training, constrain the spatial smoothness of the tracking weights, and by penalizing the spatial abrupt changes of the weights, make the weights more concentrated in the target area, reduce the influence of background noise, and enhance the tracking robustness in complex scenarios.
- We propose a mechanism for detecting temporal-spatial interference. This method uses the response value map of the previous frame to pre-detect the interference object, and builds a precise interference object screening mechanism based on the temporal response feedback. It is integrated into the training of the current frame filter to enhance the discrimination ability in complex scenarios.
- The experimental results on OTB50, OTB100 and UAV-123 show that the algorithm proposed in this paper has good robustness and accuracy in complex scenarios.

II. RELATED WORK

The target tracking method based on correlation filtering has gained extensive attention in the field of target tracking due to its advantages of high computational efficiency and good robustness. The application of correlation filtering theory to the tracking field was first proposed by Bolme et al. [8], and they put forward a MOSSE algorithm for single-target tracking. By constructing a correlation filter, the image signal is mapped from the time domain to the frequency domain, and correlation operations are performed in the frequency domain, which greatly improves the real-time performance of the model. However, the MOSSE algorithm only uses single-channel grayscale features when training filters, which cannot fully and accurately describe the features of the target in complex scenes, and the number of training samples used is also insufficient.

In 2012, Henriques et al. [9] proposed the CSK algorithm to address the issue of the number of training samples in the MOSSE algorithm. This algorithm increases the number of training samples through the method of cyclic shift sampling and introduces a kernel function to achieve rapid solution of the ridge regression problem, thereby enhancing the algorithm's performance. Although the CSK algorithm has solved the problem of insufficient training samples, it is prone to tracking drift when there are objects similar to the target in the background. To address this issue, Henriques et al. [10] proposed the KCF algorithm in 2014, which utilizes multi-channel gradient histogram features to enhance feature representation capabilities, compensating for the insufficiency of single-channel grayscale feature representation capabilities and further improving the tracking performance of the algorithm. The emergence of the KCF algorithm marks the maturation of the target tracking algorithm framework based on correlation filtering.

To alleviate the boundary effect problem, Danelljan et al. [11] proposed a spatial regularized discriminative correlation filtering (SRDCF) algorithm in 2015. By introducing a spatial regularization term into the objective function to impose targeted constraints on the boundary region, the interference generated by the boundary region during the tracking process is effectively suppressed, and the positioning accuracy is improved. However, this leads to an increase in the overall computational complexity of the algorithm. Li et al. [12] introduced the temporal regularization term on the basis of SRDCF and proposed the spatio-temporal regularized correlation Filter (STRCF) algorithm, which effectively suppressed filter degradation, simplified sample management and reduced the computational burden. Yuan et al. [13] proposed a novel visual tracking framework that combines motion perception and automatic time regularization for the fixed time regularization parameter problem of the STRCF algorithm. By incorporating the motion perception strategy based on Kalman filtering (KF) into STRCF, the problem of target loss caused by occlusion and target deformation has been effectively alleviated.

To further improve the tracking performance of the filter, Wang et al. [14] introduced the context information of the target as a negative sample into the training of the correlation filter to

enhance the background awareness ability of the model. Lin et al. [15] effectively alleviated the tracking drift problem by introducing a response-weighted background residual term to conduct targeted learning on the target background region.

III. METHODOLOGY

A. Spatiotemporal constraints

We choose the STRCF tracker as the baseline. Fig.1 shows the overall process of the algorithm proposed in this paper. The objective function defined in this article is as follows:

$$\Psi(H, \omega) = \frac{1}{2} \left\| \sum_{k=1}^K x_k^t * h_k' - y \right\|_2^2 + \frac{1}{2} \sum_{k=1}^K \left\| \omega \cdot h_k' \right\|_2^2 + \frac{\sigma}{2} \|h^t - h^{t-1}\|^2 + \|\nabla \omega\|_2^2 + \lambda \|\omega\|_1 \quad (1)$$

Where, The symbol * indicates convolution operation. $x_k^t \in R^N$ ($k = 1, 2, \dots, K$) represents the feature samples extracted in t frames, N is the size of the entire image, and K is the total number of channels in each frame of the image. $y \in R^N$ represents the ideal correlation response value, and $H = [h_1, h_2, \dots, h_K]$ and $h_k \in R^N$ are the filters of the K channel. λ is the regularization parameter. $\frac{1}{2} \sum_{k=1}^K \|\omega \cdot h_k'\|_2^2$ is a spatial

regularization term. ω is the spatial weight matrix, which makes the target area change in a gradient from the background area, thereby achieving smooth spatial constraints. $\frac{\sigma}{2} \|h^t - h^{t-1}\|^2$ is the time regularization term. σ is a coefficient and h^t and h^{t-1} represent the filters trained in the current frame and the previous frame respectively. $\|\nabla \omega\|_2^2 + \lambda \|\omega\|_1$ represents the spatial constraint term, and $\|\nabla \omega\|_2^2$ represents the spatial gradient. By constraining the sudden changes in spatial weights and making the spatial and temporal dimensions adjust in coordination, it ensures that the filter stably focuses on the trusted area of the target.

The STRCF algorithm reduces the weight of the background area by penalizing the overall amplitude of the filter weights, thereby alleviating the boundary effect. However, it cannot avoid the local fluctuations of the weights in the background noise area, and the filter is still affected by noise. Therefore, this paper introduces spatial gradient regularization. By imposing gradient constraints, it directly suppresses the spatial mutations of the weights, which can better alleviate the interference from background noise.

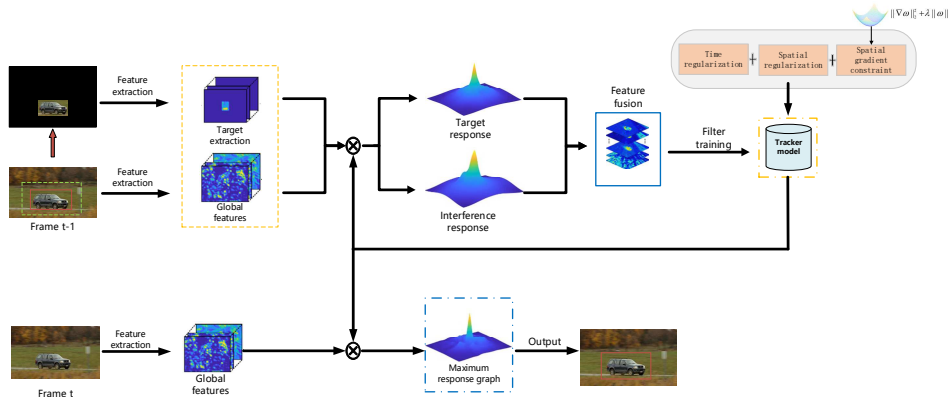


Fig.1 Overall block diagram of the system. Spatio-temporal constraint terms are added to the tracker to suppress background noise interference, and the response value maps of the target and similar distractors in the previous frame image are extracted. The features of the two are fused and added to the learning of the tracker, thereby enhancing the robustness of the tracker.

B. Optimize using the ADMM method

To enhance computational efficiency, based on Parseval's theorem, equation (1) is transformed to the frequency domain for calculation, and the auxiliary variables $\hat{g} \in C^{N \times K \times 1}$ and $\hat{G} = [\hat{g}_1, \hat{g}_2, \hat{g}_3, \dots, \hat{g}_K]$ are introduced. The transformed objective function is as follows:

$$L(H, \omega, \hat{G}) = \frac{1}{2} \left\| \sum_{k=1}^K \hat{x}_k * \hat{g}_k' - \hat{y} \right\|_2^2 + \frac{1}{2} \sum_{k=1}^K \left\| \omega \cdot h_k' \right\|_2^2 + \frac{\sigma}{2} \|h^t - h^{t-1}\|^2 + \|\nabla \omega\|_2^2 + \lambda \|\omega \cdot \sigma\|_1 \quad (2)$$

Where, $\hat{y} = \sqrt{N} P y$, $\hat{x}_k = \sqrt{N} P x_k$, and $\hat{g}_k = \sqrt{N} P g_k$ are Fourier transforms, and $P \in C^{N \times N}$ is an orthogonal matrix, which can map N -dimensional vector signals to the Fourier domain.

To find the optimal solution by minimizing the formula using the ADMM method, equation (2) can be transformed into an augmented Lagrangian form:

$$L(H, \omega, \hat{G}, \hat{D}) = E(H, \omega, \hat{G}) + \sum_{k=1}^K \hat{d}_k (\hat{g}_k - \sqrt{N} P h_k)^T + \frac{\rho}{2} \sum_{k=1}^K \left\| \hat{g}_k - \sqrt{N} P^T h_k \right\|_2^2 \quad (3)$$

Where, $\hat{D} = [\hat{d}_1, \hat{d}_2, \hat{d}_3, \dots, \hat{d}_K] \in R^{N \times K}$ is the Lagrange multiplier. ρ is the ADMM penalty factor. Let $m_k = \frac{1}{\rho} \hat{d}_k$, equation (3) be transformed into:

$$L(H, \omega, \hat{G}, \hat{D}) = E(H, \omega, \hat{G}) + \frac{\rho}{2} \sum_{k=1}^K \left\| \hat{g}_k - \sqrt{N} P^T h_k + m_k \right\|_2^2 \quad (4)$$

By using the ADMM algorithm, the equation can be

decomposed into the solutions of three sub-problems: H , $\hat{G}$, and ω .

a. Solve for H

Suppose given ω , $\hat{G}$ and $\hat{D}$, the objective function is:

$$L(h_K) = \frac{\rho}{2} \left\| \hat{g}_K - \sqrt{N} P^T h_K + m_K \right\|_2^2 + \frac{\sigma}{2} \left\| \omega \cdot h_K \right\|_2^2 \quad (5)$$

Let $\frac{\partial L(h_K)}{\partial h_K} = 0$ solve the above equation, and we get:

$$h_K^* = \frac{\rho N (g_K + m_K)}{\sigma (W \cdot W) + \alpha N} \quad (6)$$

Where, $W = \text{diag}(\omega) \in R^{N \times N}$ represents the diagonal matrix of the spatial regular weights.

b. Solve for spatial regular weights ω

Suppose given H , $\hat{G}$ and $\hat{D}$, the objective function is:

$$L(\omega) = \frac{1}{2} \sum_{k=1}^K \left\| \omega \cdot h_k^t \right\|_2^2 + \left\| \nabla \omega \right\|_2^2 + \lambda \left\| \omega \right\|_1 \quad (7)$$

Introduce auxiliary variables z_k and u . Let $z_k = \omega \cdot h_k^t$ ($k=1,2,K,K$) and $u = \nabla \omega$ transform equation (7) into:

$$L(\omega) = \frac{1}{2} \sum_{k=1}^K \left\| z_k \right\|_2^2 + \left\| u \right\|_2^2 + \lambda \left\| \omega \right\|_1 \quad (8)$$

The augmented Lagrangian form of Equation (8) can be expressed as:

$$L(\omega, q, \alpha, \beta) = \psi(\omega) + \frac{\rho}{2} \left\| u - \nabla \omega + \beta \right\|_2^2 + \frac{\rho}{2} \sum_{k=1}^K \left\| z_k - \omega \cdot h_k^t + \alpha_k \right\|_2^2 \quad (9)$$

Where, α_k and β are Lagrange multipliers.

Take the partial derivative of equation (9) and set it equal to 0 to obtain the closed-form solution:

$$\left(2 \nabla^T \nabla + \rho \sum_{k=1}^K h_k^t (h_k^t)^T + \rho I \right) \omega = \rho \sum_{k=1}^K h_k^t (z_k + \alpha_k) + \rho \nabla^T (u + \beta) \quad (10)$$

Where, $\nabla^T \nabla$ is the Laplacian matrix, I is the identity matrix, and ∇^T is the gradient transpose operator.

Equation (10) is split into the solutions of two optimization sub-problems, z_k and u , and the updates of α_k and β , through the ADMM algorithm, and the solutions are iteratively carried out step by step.

The optimization subproblems for z_k and u are

$$\begin{cases} z_k = \min \frac{1}{2} \left\| z_k \right\|_2^2 + \frac{\rho}{2} \left\| z_k - \omega \cdot h_k^t + \alpha_k \right\|_2^2 \\ u = \min \left\| u \right\|_2^2 + \frac{\rho}{2} \left\| u - \nabla \omega + \beta \right\|_2^2 \end{cases} \quad (11)$$

The closed-form solutions and iterative forms of the two

optimization sub-problems are

$$\begin{cases} z_k = \text{sign}(\omega \cdot h_k^t - \alpha_k) \max \left(\left| \omega \cdot h_k^t - \alpha_k \right| - \frac{1}{2\rho}, 0 \right) \\ u = \frac{\rho}{2 + \rho} (\nabla \omega - \beta) \end{cases} \quad (12)$$

The iterative updates of α_k and β in Equation (12) are respectively

$$\alpha_k^{j+1} = \alpha_k^j + z_k^{j+1} - z_k^j \quad (13)$$

$$\beta^{j+1} = \beta^j + u^{j+1} - u^j \quad (14)$$

c. Solve for $\hat{G}$

Suppose given H , ω and $\hat{D}$, the objective function is:

$$L(\hat{G}) = \frac{1}{2} \left\| \sum_{k=1}^K \hat{x}_k * \hat{g}_k^t - \hat{y} \right\|_2^2 + \frac{\alpha}{2} \sum_{k=1}^K \left\| \hat{g}_K - \sqrt{N} P^T h_K + m_K \right\|_2^2 \quad (15)$$

Due to the existence of the summation operation, the direct computational complexity of differentiating $\hat{g}_k$ is relatively high, making it difficult to optimize the operation. To reduce the complexity, it can be considered to process on all channels of each pixel. Let $\psi_i(\hat{G}) \in R^K$ represent the vector of all K channel values on pixel i, which can be transformed into:

$$L_\psi(\hat{G}) = \frac{1}{2} \sum_{k=1}^K \left\| \psi_i(\hat{X}) \psi_i(\hat{G}) - \hat{y}_i \right\|_2^2 + \frac{\alpha}{2} \sum_{k=1}^K \left\| \psi_i(\hat{G}) - \psi_i(\sqrt{N} P H) + \psi_i(m_K) \right\|_2^2 \quad (16)$$

Solving for:

$$\psi_i(\hat{G}) = \frac{\psi_i(\hat{X}) \hat{y}_i + \alpha N [\psi_i(\hat{M}) - \psi_i(\sqrt{N} P H)]}{\psi_i(\hat{X})^T \psi_i(\hat{X}) + \alpha N} \quad (17)$$

In the iterative steps of ADMM, the Lagrange multiplier $\hat{D}$ is updated according to the following formula:

$$\hat{D}^{j+1} = \hat{D}^j + \hat{G}^{j+1} - \hat{H}^{j+1} \quad (18)$$

Where, j represents the number of iterations, and $\hat{D}^j$ represents the Fourier transform of the Lagrange term in the j -th iteration.

C. Detection of time-space interference objects

Traditional CF only utilize the information of the current frame and thus have difficulty in distinguishing the target from similar interfering objects. To address this issue, researchers have introduced the background-aware modeling method into target tracking, enhancing the discriminative ability of the tracker. However, by only using the background region of the current frame and not combining historical tracking information, it is impossible to capture the dynamically emerging interfering objects, and the predefined background

region may introduce invalid samples, resulting in ineffective learning of the tracker. Therefore, this paper introduces the spatio-temporal interference detection method, using the response value map of the previous frame to pre-detect interfering objects, and based on the temporal response feedback, constructing an accurate interference object screening mechanism, forming a "target-interfering object" dual learning paradigm, and integrating it into the training of the current frame filter to enhance the discrimination ability in complex scenarios.

The response value graph is the quantitative result of the filter for the "similarity to the target" of the image area. The higher the response value, the stronger the appearance consistency between the area and the target. The position corresponding to the point with the highest response value is regarded as the center point of the predicted target.

The calculation formula for the response value graph R:

$$R_t = \mathcal{F}^{-1}(\hat{h}^{t-1} \odot \hat{x}^{t-1}) \quad (19)$$

Where, $\mathcal{F}^{-1}$ represents the inverse Fourier operation, $\odot$ represents the dot product operation, $\hat{h}^{t-1}$ represents the filter, and $\hat{x}^{t-1}$ represents the target appearance model.

The high-response region is a candidate set of distractors. To avoid introducing too many low-similarity regions, threshold screening is required. Moreover, due to the movement of the target and the lens during the tracking process, the detected response values often change significantly. Therefore, a dynamic threshold θ is needed to determine the distractors.

$$\theta = \delta \odot \max(R_{t-1}) \quad (20)$$

Where, δ is the threshold coefficient and $\delta \in (0,1)$. It can be adaptively adjusted according to the complexity of the scene.

Select the regions where the response values are greater than the threshold and denote them as candidate set S_{date} .

$$S_{date} = \{(i, j) | R_{t-1}(i, j) > \theta\} \quad (21)$$

Since candidate set S_{date} may contain the target area of the previous frame, it needs to be excluded through the intersection of union (IOU).

$$IOU(B_{t-1}, B_k) = \frac{Area(B_{t-1} \cap B_k)}{Area(B_{t-1} \cup B_k)} \quad (22)$$

Where, B_{t-1} represents the target box of the previous frame, and B_k represents the candidate box.

Set the overlap threshold τ and retain the candidate boxes of $IOU < \tau$, that is, exclude the regions with a high degree of overlap with the target. The remaining regions are denoted as the sets of interfering object regions S_{dist} , $S_{dist} = \{B_k | IOU(B_{t-1}, B_k) < \tau\}$. In this article, the value of τ is 0.3.

For each region in the set of distractor regions, extract features consistent with the target features, merge the target samples with the distractor samples, and form an extended training set.

IV. RESULT AND DISCUSSION

To ensure the fairness of the algorithm, all experiments in this paper were conducted on a computer platform with Windows 11 operating system, Intel(R) Core(TM) i5-9500 CPU, 3.00GHz and 8G memory using Matlab2021b software for simulation experiments. We conducted a comprehensive evaluation of the proposed tracker on three popular visual tracking benchmarks: OTB50, OTB100 and UAV-123. We train the tracker through 2 ADMM iterations and the learning rate η is set to 0.014. The other parameters are $\lambda = 0.005$, $\sigma = 0.05$.

This article will conduct a comparative analysis from both qualitative and quantitative aspects.

A. Quantitative analysis

The OTB50 dataset. This dataset includes 11 different challenging factors, thus serving as the comprehensive performance evaluation criterion for the algorithm in this paper. Five advanced algorithms were selected to be compared with the algorithm in this paper on the dataset OTB50. The five comparison algorithms are: STRCF [10], SRDCF [11], ECO_HC [16], and DSST [17]. The experimental results are shown in Fig.2.

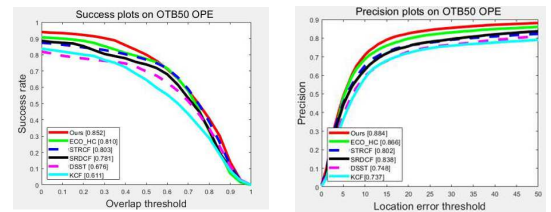


Fig.2 The success rate and accuracy rate of each algorithm on the OTB50 dataset.

As shown in Fig.2, the algorithm in this paper achieved scores of 0.852 for success rate and 0.884 for accuracy, respectively, which were superior to the benchmark algorithm STRCF (6.1% and 10.2%). Our method achieved a running

speed of 47.4 FPS, which is close to the corresponding STRCF (49.7 FPS). This verifies the high efficiency of our method.

TABLE I. A COMPARISON WITH FIVE ADVANCED TRACKERS ON THE OTB50 BENCHMARK

Algorithm	Measurement indicators		
	Success rate	Accuracy rate	FPS
Ours	0.852	0.884	47.4
ECO_HC	0.810	0.866	41.6
STRCF	0.803	0.802	49.7
SRDCF	0.781	0.838	5.7
DSST	0.676	0.748	37.8
KCF	0.611	0.737	232.6

The OTB100 dataset. The OTB100 dataset consists of 100 challenging video sequences and is an extension of OTB-50. The dataset includes various scenarios such as motion blur, occlusion, and scale variation. Five trackers, namely STRCF, BACF[18], AMCF[19], MUSTER[20], and SAMF[21], were selected for comparison with this algorithm. The experimental results are shown in Fig.3:

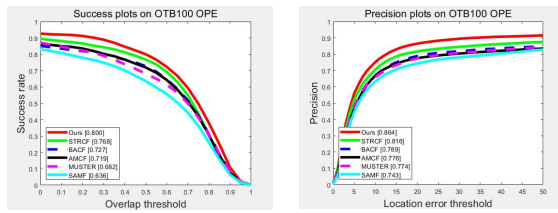


Fig.3 The success rates and accuracy rates of various algorithms on the OTB100 dataset.

TABLE II. A COMPARISON WITH FIVE ADVANCED TRACKERS ON THE OTB100 BENCHMARK

Algorithm	Measurement indicators		
	Success rate	Accuracy rate	FPS
Ours	0.800	0.864	25.1
STRCF	0.768	0.816	27.8
BACF	0.727	0.789	37.7
AMCF	0.719	0.776	28.2
MUSTER	0.682	0.774	5.4
SAMF	0.636	0.743	56.1

In Fig.3, the overall success rate of our method on the OTB100 dataset is 0.800, and the accuracy rate is 0.864. Compared with the success rate (0.768) and accuracy rate (0.816) of the benchmark algorithm STRCF, it has increased by 4.2% and 5.9% respectively.

The UAV-123 dataset. The UAV-123 dataset is a widely used benchmark captured by unmanned aerial vehicles. To conduct a more comprehensive evaluation of the method proposed in this paper, we further evaluated the tracking performance on the UAV-123 dataset. We introduced spatial

gradient suppression and spatio-temporal interference detection mechanisms in DeepSTRCF and conducted comparative experiments with trackers STRCF, DeepSTRCF[22], SiamRPN[23], SiamRPN++[24], and LADCF_HC[25]. The experimental results are shown in Fig.4.

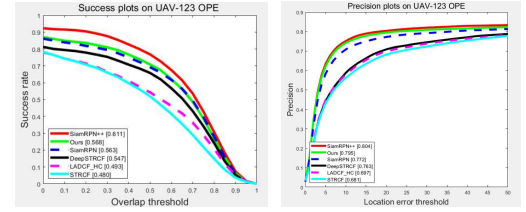


Fig.4 The success rate and accuracy rate of each algorithm on the UAV-123 dataset.

TABLE III. A COMPARISON WITH FIVE ADVANCED TRACKERS ON THE UAV-123 BENCHMARK

Algorithm	Measurement indicators	
	Success rate	Accuracy rate
SiamRPN++	0.611	0.804
Ours	0.568	0.795
SiamRPN	0.563	0.772
DeepSTRCF	0.547	0.763
LADCF_HC	0.493	0.697
STRCF	0.480	0.681

As shown in Table III, in terms of success rate, our tracker achieved a score of 0.568, which was 3.8% and 18.3% higher than the benchmark DeepSTRCF (0.547) and STRCF (0.480), respectively.

B. Qualitative analysis

Fig.5 shows video sequences with 11 different attributes such as occlusion, deformation, and illumination variation in the OTB100 dataset. It can be seen from the figure that the algorithm proposed in this paper has better tracking performance compared with the other five algorithms.



(a) Woman tracking schematic diagram



(b) Trans tracking schematic diagram



(c) Waiking2 tracking schematic diagram

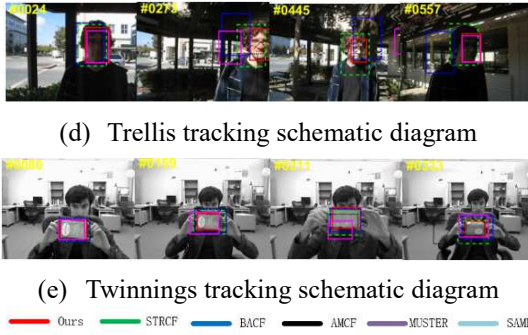


Fig.5 Comparison of the tracking results of six algorithms in the test video

In the Woman and Waiking2 sequences, when the target is partially occluded or subject to motion blur, the STRCF algorithm loses the target, while the method proposed in this paper can always accurately locate the target position. This is because the method in this paper introduces spatial gradient regularization and temporal-spatial interference detection strategies, combined with the information of historical filters, significantly alleviates the interference of the background, and accurately locates the target. In the sequence Twinnings, from frame 159 to 333, the target undergoes two consecutive rotations causing deformation. Apart from the tracker in this paper being able to precisely track the target, other algorithms can only track a part of the target. In the sequences Trans and Trellis, the background illumination changes, and STRCF exhibits tracking drift phenomenon. However, the method proposed in this paper successfully locates the target under background interference. In summary, our method has better tracking performance than STRCF in complex situations such as scale transformation, deformation, rapid movement, rotation, and occlusion.

C. Ablation experiment

We conducted ablation experiments on the OTB100 dataset to verify the efficiency of the key components in the proposed method. The basic concepts are as follows. (1) "STRCF + SCT" refers to the addition of a spatial gradient regularization mechanism. (2) "STRCF + TD" indicates the STRCF with the introduction of a temporal-spatial interference object detection module. (3) "STRCF + SCT + TD" is our final tracker that combines STRCF with the spatial gradient regularization mechanism and the temporal-spatial interference object detection module. The tracking results are shown in Fig.6.

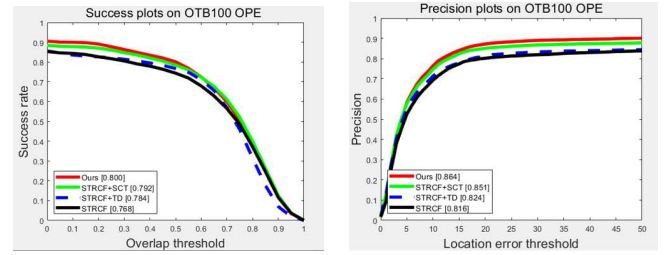


Fig.6 Ablation experiments using the OTB100 dataset

TABLE IV. THE RESULTS OF THE ABLATION EXPERIMENT

Algorithm	Measurement indicators		
	Success rate	Accuracy rate	FPS
STRCF	0.768	0.816	27.8
STRCF+SCT	0.792	0.851	25.3
STRCF+TD	0.784	0.824	22.5
STRCF+SCT+TD(Ours)	0.800	0.864	25.1

In Fig.6, we observe that the success rates of "STRCF+SCT" and "STRCF+TD" have increased by 3.1% and 2.1% respectively compared to the baseline algorithm, and the accuracy has improved by 4.2% and 1% respectively. Therefore, the spatio-temporal constraint and spatio-temporal interference detection mechanism proposed in this paper can significantly enhance the performance of STRCF. Additionally, the accuracy and success rate of the algorithm in this paper have increased by 5.9% and 4.2% respectively compared to the baseline algorithm.

D. Dynamic threshold sensitivity analysis

The improvement of the time-space interference detection module involves the overlapping threshold parameter. This parameter is used to define the interference object and the target. Only when $\text{IOU} < \tau$ will it be included in the negative sample training, avoiding mistakenly treating the target itself or the overlapping area as an interference object.

TABLE V. TRACKING PERFORMANCE UNDER DIFFERENT THRESHOLDS IN THE OTB100 DATASET

Threshold τ	0.2	0.3	0.4	0.5
Success rate	0.786	0.800	0.784	0.761
Accuracy rate	0.847	0.864	0.806	0.788

As can be seen from Table V, when the threshold is set to 0.3, the performance of our tracker reaches its peak.

V. CONCLUSIONS

This paper proposes a spatio-temporal constrained anti-interference correlation filtering target tracking algorithm. This

algorithm introduces a spatial gradient regularization term in the spatial constraint module of the STRCF algorithm, constraining the spatial smoothness of the tracking weights and suppressing background noise interference. Additionally, in view of the influence of similar interferers on the tracking robustness, by pre-detecting the response value graph of the previous frame, it looks for possible interferer information similar to the target and incorporates it into the training of the filter. Improve the discrimination ability of the filter. The algorithm was evaluated on the public dataset OTB50, OTB100 and UAV-123. The experimental comparison results show that the algorithm proposed in this paper can effectively alleviate the problem of poor tracking robustness in abnormal situations such as deformation, occlusion, and rotation, and also performs well in terms of success rate and accuracy.

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